

PANAGIOTIS OIKONOMOPOULOS

Pentester – Red Teamer

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HTB PROFILE

My writeup blog



SUMMARY

Electrical and Computer Engineering student specializing in Offensive Security and Red Teaming. Experienced in penetration testing, exploit development, and adversary simulation across enterprise, cloud, and web environments. Developed custom offensive tooling including polymorphic shellcode loaders and AWS auditing frameworks, and built virtualized labs using pfSense and Wazuh to simulate and analyze real-world attacks. Passionate about evasion techniques, reverse engineering, and breaking systems to strengthen their security.

EDUCATION

University of the Peloponnese

Electrical and Computer Engineer

SKILLS

- **Offensive Security** – Red team operations, evasion techniques, C2 frameworks (Cobalt Strike, Sliver)
- **Tool Development** – Custom malware, shellcode loaders, Python/C++ offensive tooling
- **Penetration Testing** – AD, Network, Web, AWS, GCP, AZURE and Android application.
- **Programming** – Python, C++, Windows API
- **Threat Analysis** – pfSense, Wazuh, Wireshark
- **Cryptography** – Encryption, secure communications
- **Collaboration** – Experience in technical team environments
- **Reverse Engineering**: Frida, IDA, Jadx, x86-64

CERTIFICATIONS

CYBERSECURITY

- CPTS | Hack The Box
- CRTL | Zero-Point Security
- CRTO | Zero-Point Security
- PIPT | TCM Security
- Cloud Master | Extreme Red Team Laboratories
- Bug Bounty Hunter | HTB
- Penetration Tester | HTB
- Zephyr ProLab | HTB
- FullHouse ProLab | HTB
- Dante ProLab | HTB
- P.O.O ProLab | HTB
- AI Red Teamer | HTB
- Jr Penetration Tester | THM
- FCA | Fortinet
- FCF | Fortinet
- AWS Pentest | SimplyCyber
- Windows API for Red Team | CourseStack
- MCRTA | Cyberwarfare
- CRTA | Cyberwarfare
- AVUESSO | Extreme Red Team Laboratories
- CYAN | Extreme Red Team Laboratories
- STORM | Extreme Red Team Laboratories
- SUMMUS | Extreme Red Team Laboratories
- Red Teaming | THM
- Offensive Pentesting | THM

PROGRAMMING

- Database Design | Oracle Academy
- Database Programming with SQL | Oracle Academy
- Meta Front-End Developer | Meta
- Computational Motion Planning | Penn University
- Aerial Robotics | Penn University

PROFESSIONAL EXPERIENCE

Cloud & Cybersecurity Consultant

OTS | 2025 – Present

- **Conducted** penetration tests on enterprise environments, delivering professional reports to enhance security measures.
- **Designed** and led phishing simulation campaigns to improve organizational security awareness.
- **Developed** NIS2 compliance reports and automations, achieving a 99% reduction in creation time.
- **Executed** thorough vulnerability assessments utilizing tools like Nessus and OpenVAS to identify security gaps.

Cybersecurity Intern

ShadowFox | 1 month 2025

- **Learned** - Hands-on experience in cybersecurity, work on real-world projects, certification, and enhancement of professional profile.

Cybersecurity Intern

The Red Users | 1 month 2025

- **Learned** - Practical skills in cybersecurity, tackling real-world problems, exploring the field of web application security.

Software Engineer

Neux Ltd | 2023– 2024

- **Frontend Development for XR Applications** – Creating interactive experiences in virtual and augmented reality environments.
- **Bot Development** – Automating processes and enhancing user interaction across various platforms.
- **API Implementation and Integration** – Developing APIs for system communication and seamless application functionality.

Lead Software Engineer

Hyperion Drone Team | 2023 – Present

- **PID Controller Development and Optimization** – Designing and tuning PID controllers for drone flight control, ensuring stability and precise responsiveness.
- **AI Model Creation and Training for Object Detection** – Enhancing drone autonomy and navigation capabilities through trained AI object detection models.
- **Flight System Simulations** – Using Matlab for performance analysis and evaluation of the flight system through simulations.
- **Custom Flight Controller Design and Development** – Building a tailored flight controller to meet the specific needs of the team and drone applications.

PROJECTS

Capstone Project – Advanced Shellcode Obfuscation & AV Evasion Techniques

- Developed a multi-version polymorphic shellcode loader using C++ with progressive evasion strategies targeting Windows Defender.
- Implemented XOR encryption and polymorphic obfuscation, gradually advancing to indirect syscall execution and API hashing (MurmurHash3) to avoid AV detection.
- Demonstrated evasion improvement across six versions, culminating in complete bypass of Windows Defender in Version 4.
- Tested shellcode delivery through custom encoders and decoders, integrating runtime stubs and randomized key generation.
- Applied cybersecurity principles and red team tactics, including payload packing, memory protection changes, and antivirus sandbox behavior analysis.

More info about the project its on my [**blog**](#).

Network Security Lab – Firewall, XDR & Traffic Analysis

- Designed and deployed a virtualized network lab to explore foundational network security using VirtualBox, pfSense, and Wazuh.
- Configured pfSense firewall/router with NAT and internal LAN/WAN segmentation to simulate realistic enterprise boundaries.
- Installed Wazuh (XDR/SIEM) server on Ubuntu and agent on LinuxMint to detect and log suspicious behavior through command execution simulations.
- Enforced firewall rules to block external ICMP traffic and verified network isolation and routing through controlled connectivity tests.
- Monitored traffic using Wireshark to analyze DNS queries, XDR heartbeat signals, and ICMP filtering effectiveness.

More info about the project is available on my [**blog**](#).

CloudTap an AWS IAM & S3 Auditor – Python Security Tooling

- Developed a Python script using Boto3 to audit AWS IAM users, groups, roles, and policies.
- Enumerated attached and inline policies for IAM users and groups, printing permissions in readable JSON format.
- Integrated functionality to download files from S3 buckets and retrieve secrets from AWS Secrets Manager.
- Parsed and analyzed trust policies to identify potentially assumable IAM roles based on the caller's identity.
- Included support for optional S3 downloads, role assumption testing, and structured secret inspection using custom JSON serializers.

More info about the project is available on my [**github**](#).

Drone Ground Station – Secure Long-Range Communication System

Hyperion UAV Team | 2025

- Collaborating with the Hyperion drone team to develop a custom ground station enabling secure two-way communication and control of UAVs over distances up to 10km.
- Built a modular system integrating a web-based user interface for drone telemetry, diagnostics, and remote task execution.
- Integrated embedded AI subsystems to enhance autonomy, responsiveness, and real-time decision support.
- Focused on system-level security, communication reliability, and performance under field conditions.

Further technical details are confidential due to project sensitivity.